MATH 340 Examination

Multivariable Calculus and Vector Analysis – Answer Key

1. (C) The direction of steepest ascent and perpendicular to level sets.
The gradient ∇f points in the direction where f increases most rapidly, and its magnitude equals the rate of that increase. It is always perpendicular to the level sets (curves/surfaces where f is constant).

2. (C) f has continuous second partial derivatives.
Schwarz's theorem (Clairaut's theorem) states that if the second partial derivatives f_{xy} and f_{yx} are continuous in a neighborhood of a point, then they are equal at that point. Continuity of second partials is sufficient; mere existence is not.

3. (B) The local scaling factor of volumes under the transformation.
The Jacobian determinant $|\det(J_T)|$ measures how the transformation T locally stretches or compresses volumes. It appears as the scaling factor in the change of variables formula for multiple integrals.

4. (A) Green's Theorem is a special case of Stokes' Theorem in two dimensions.
Stokes' Theorem relates the integral of curl over a surface to the line integral around its boundary. Green's Theorem is the 2D version where the surface is flat (in the xy-plane) and curl reduces to $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

5. *Sample answer:*
The Hessian matrix H_f is the matrix of second partial derivatives:

$$H_f = \begin{bmatrix} f_{x_1x_1} & \cdots & f_{x_1x_n} \\ \vdots & \ddots & \vdots \\ f_{x_nx_1} & \cdots & f_{x_nx_n} \end{bmatrix}$$

Geometric significance: The Hessian captures the curvature of f at a critical point.

Second derivative test:

- If H_f is positive definite (all eigenvalues positive), the critical point is a local minimum
- If H_f is negative definite (all eigenvalues negative), it's a local maximum
- If H_f has both positive and negative eigenvalues, it's a saddle point
- If H_f is singular or indefinite but not clearly positive/negative, the test is inconclusive

6. *Sample answer:*
Conservative field: A vector field $\mathbf{F}$ is conservative if $\mathbf{F} = \nabla f$ for some scalar potential function f.

Path-independent field: $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on endpoints, not on the path C.

Equivalence: In simply connected domains, these are equivalent. A field is conservative if and only if it is path-independent.

Role of curl: If $\mathbf{F}$ is conservative, then $\nabla \times \mathbf{F} = \mathbf{0}$ (curl is zero). Conversely, in simply connected regions, if curl is zero, then $\mathbf{F}$ is conservative. The curl being zero is a necessary local condition for conservativeness.

7. *Sample answer:*
Divergence Theorem: For a solid region E bounded by surface S with outward normal $\mathbf{n}$:

$$\iiint_E (\nabla \cdot \mathbf{F})\, dV = \iint_S \mathbf{F} \cdot \mathbf{n}\, dS$$

Physical interpretation: In fluid flow, $\nabla \cdot \mathbf{F}$ measures the net rate of fluid expansion per unit volume (sources minus sinks). The theorem states:

- The total outward flux through the boundary equals the total divergence inside
- If there are sources inside E, net flux is outward; if sinks, inward
- It connects local property (divergence at each point) to global property (total flux)

This is fundamental in electromagnetism (Gauss's law) and fluid dynamics.

8. *Sample answer:*
When changing order of integration, the limits must reflect the same region R:

- **Method**: Sketch the region R in the xy-plane
- For $\int_{x=a}^{b} \int_{y=g_1(x)}^{g_2(x)} f(x,y)\, dy\, dx$, first identify the vertical cross-sections
- To switch to $\int\int dx\, dy$, consider horizontal cross-sections: for each fixed y, determine which x values lie in R
- Express x as a function of y: find $h_1(y) \le x \le h_2(y)$ and the range of y

Example: For R bounded by $y = x^2$ and $y = 1$, switching from $\int_0^1 \int_{x^2}^1 dy\, dx$ to $dy\, dx$ order gives $\int_0^1 \int_0^{\sqrt{y}} dx\, dy$.